WEEK 2 OF 5 Â· STRATEGY DELIVERABLE

Data Cleaning and Preprocessing Strategy

How a deliberately messy agribusiness panel is profiled, repaired, documented and made reproducible — with a ten-step cleaning plan, a five-layer preprocessing framework and a worked reference pipeline.

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TEST PANEL
3,456 price cells
6 crops Â· 8 states Â· 72 months

DEFECT CLASSES
Eight
with planted, measurable defects

EVIDENCE
550 log rows
from a seeded reference pipeline

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Document control

Table 1. *Document identification and control record.*

Item	Detail
Title	Data Cleaning and Preprocessing Strategy
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1. Executive Summary

This document sets out how a messy agribusiness dataset is turned into an analysis-ready panel without quietly changing what the data says. It is written for the dataset described in Week 1: six crops, eight Indian states, monthly wholesale prices and annual yields, with weather and trade series attached later. The strategy is generic where it can be and specific where specificity prevents error — unit conversion, geography mapping and vintage handling are all agricultural problems before they are statistical ones.

The central claim is simple. Cleaning is not a preliminary chore that happens before the analysis; it is the part of the analysis that determines whether the later results mean anything. A forecast built on a price series that silently mixes rupees per kilogram with rupees per quintal will have a plausible shape and a wrong level. A yield model trained on revised production estimates that were not available at the forecast origin will look accurate and be unusable. The methods below are designed to make those failures visible, measurable and hard to repeat.

The strategy rests on five commitments:

- **One shape —** one canonical grain, one canonical unit and one canonical key per fact, defined before any cleaning begins.
- **Classify, then repair —** a gap caused by a market holiday is not the same as a gap caused by a missing file, and neither is the same as a value withheld by the publisher. Treatment follows cause.
- **Rules, not edits —** every intervention is a named, parameterised function with a test, not an edit in a notebook. The rule can be reviewed, versioned and re-run.
- **Measure the change —** the count of corrected, imputed, flagged and dropped values is published beside the clean table, so the effect of cleaning is part of the result rather than hidden behind it.
- **Rebuild or it is not done —** a clean clone, a pinned environment and a sealed seed reproduce the panel hash exactly. If they do not, the pipeline is not finished.

1.1 What the reference implementation demonstrates

The strategy is not left as an intention. A seeded generator builds the hypothetical Mandi–Yield Panel 2026 with known defects planted at documented rates, and a reference pipeline applies the cleaning rules in order. The report quotes the pipeline's own output, not numbers typed into the prose. The headline result is small enough to state plainly: usable price cells rise from 92.3% to 98.5%, duplicate business keys fall from 2.2% to zero, and every one of the 550 logged interventions carries a rule identifier and a reason.

COMPLETENESS	DUPLICATES	IMPUTED	LOG ROWS
98.5%	**0**	**4.5%**	**550**
from 92.3% before cleaning	79 rows removed on the business key	156 cells, all gaps of 3 months or less	9 rules fired; every change attributable

Two results deserve emphasis because they cut against the instinct to clean aggressively. First, 41 zero prices were reclassified as closed markets rather than imputed: a zero is information about trading, not a missing number, and filling it would invent a transaction that never happened. Second, 1 spike was flagged and deliberately retained — a documented price event that a naive outlier filter would have deleted, taking the most informative part of the series with it.

> **THE STRATEGY IN ONE SENTENCE**
>
> Freeze the raw data, classify every defect by cause, repair only what can be repaired with evidence, flag the rest, log every intervention, and prove the whole chain rebuilds from a clean clone.

1.2 How this document is organised

Table 2. *Document map. Sections 5–7 carry the method; the appendices carry the operational detail.*

Part	Sections	Question answered
Problem	2–4	What does the data look like, how does it fail, and what rules govern the response?
Method	5–7	What exactly is done, in what order, with which method and why?
Evidence	8–10	What does the sequence look like, how is it reproduced, and what happens to six real cases?
Delivery	11–12	How is quality signed off, and how does the work land in five days?
Reference	A–H	Schemas, registers, checklists, glossary, statistical notes and tests.

The document assumes no particular software stack, but the reference pipeline uses Python, pandas and NumPy. Nothing in the method depends on that choice. Where a rule needs a threshold, the threshold is stated with the reasoning behind it, so that a team using a different tool can still implement the same decision.

2. The Hypothetical Dataset

A strategy document that discusses missing values in the abstract is easy to write and hard to test. This one works against a defined panel, so every rule can be demonstrated and every claim measured. The panel is hypothetical, but its defects are modelled on the ones that actually appear in Indian agricultural statistics: Agmarknet price pulls, DES production estimates, and the state and district nomenclature that changes around them.

2.1 Panel design

The Mandi–Yield Panel 2026 (MYP-26) is a crop–state–month price panel with an annual production and yield panel attached. The grain is deliberate: monthly state-level modal price for the price table, annual crop–state record for the yield table. Daily mandi quotes sit below this grain and are aggregated up; district yields sit below it and are aggregated with a ratio of sums. Fixing the grain first is what makes the duplicate key unambiguous.

Table 3. *MYP-26 at a glance. The clean price panel is a complete 6 × 8 × 72 grid.*

Property	Price panel	Yield panel
Grain	crop × state × month	crop × state × year × release
Crops	6 — wheat, paddy, maize, soybean, cotton, onion	same six
States	8 — Maharashtra, Madhya Pradesh, Punjab, Haryana, Uttar Pradesh, Karnataka, Gujarat, Rajasthan	same eight
Period	January 2020 – December 2025, 72 months	2020–2025, six years
Rows	3,456 clean rows; 3,535 raw rows	480 rows across 10 release dates
Key	crop_code + state_lgd + period	crop_code + state_lgd + crop_year + vintage_id
Target use	monthly modal price, 1/3/6/12-month horizons	annual yield, t/ha

The clean grid is 3,456 cells — six crops, eight states, seventy-two months. That completeness is a design decision, not an accident: a full grid makes a missing cell an explicit row with a null and a flag, rather than a row that silently does not exist. Row-absence is the most expensive kind of missingness in a panel, because every join after it quietly loses observations and no null count ever reveals the loss.

2.2 Schema

Every fact table carries the same control columns. The schema is the contract that makes the cleaning log joinable back to the data: a log entry names a row, a field, an old value and a new value, and the row identifier is stable across the run.

Table 4. *Core fields. The full dictionary extends this with crop-year, unit and indicator fields.*

Field	Type	Example	Purpose
row_id	text	P02786	Stable identifier for the raw row; the log's foreign key
crop	text	MAIZE	Canonical crop code after harmonisation
state	text	Maharashtra	Canonical state name; LGD code in the production build
month	date	2023-07-01	ISO month start; never a formatted string
modal_price_inr_quintal	float	2,410.00	Canonical unit: rupees per quintal
source_id	text	AGMARKNET	Foreign key to the source register
vintage_id	text	2026-10-27	Publication date of the release used
retrieved_at	timestamp	2026-10-27T09:14:00Z	When the snapshot was taken
quality_flag	enum	imputed_short_gap	Never null; drives model inclusion
outlier_z	float	1.84	Robust z on log price; diagnostic, not a filter

2.3 How the defects are planted

The generator injects defects at known rates and records what it did in a manifest that the cleaning pipeline never reads. That separation matters. If the pipeline were told where the errors were, it could achieve a perfect score by looking up the answer, and the recovery rates in this report would be worthless. The pipeline has to find the defects with the same rules it would use on real data.

Table 5. *Planted defects and their rates. Every rate is computable from the manifest, not assumed.*

Defect	Mechanism	Rate	Cells / rows
Duplicate retrieval	Same business key re-pulled and appended	2.23%	79 rows
Missing price	Cell blank in the source extract	4.80%	166 cells
Zero price	Market closed or no trade	1.19%	41 cells
Unit drift	Rupees per kg recorded under a per-quintal label	1.10%	38 cells
Decimal slip	One-place transcription error	0.90%	31 cells
Digit error	A digit changed with no clean factor	0.20%	7 cells
Date format	DD/MM/YYYY and Mon-YYYY mixed into ISO	3.39%	117 rows
Crop alias	Paddy(Dhan), Rice, Soyabean, Wheat FAQ	2.00%	69 rows
State alias	Maharastra, MP, U.P., Gujrat	1.50%	52 rows
Negative price	Impossible value from a sign error	0.17%	6 cells
Production unit	Tonnes filed as thousand tonnes	1.25%	6 rows
Area unit	Acres filed as hectares	0.83%	4 rows
Missing yield	Yield not reported for a record	1.04%	5 rows

The defects are not independent. A unit-drift cell that also falls in a missing block is only a missing cell; a duplicate row that also received an alias is a duplicate only after the alias is resolved. That overlap is realistic, and it is why the rules have a fixed order. Applying outlier detection before unit repair, for example, would flag perfectly valid prices as extreme because they are expressed in a different unit.

2.4 Generator extract

The generator is short enough to show the mechanism. The point is not the price model; it is that every defect is written deliberately, with a recorded index, so the strategy can be scored against ground truth that the cleaner cannot see.

```
rng = np.random.default_rng(20261104)          # sealed seed, reported in the manifest
idx = take(38)                                 # 38 unit-drift cells
df.loc[idx, "modal_price"] = df.loc[idx, "modal_price"] / 100.0
log["rates"]["unit_drift"] = len(idx) / n

idx = take(31)                                 # 31 one-place slips
df.loc[idx[:21], "modal_price"] *= 10
df.loc[idx[21:], "modal_price"] /= 10

manifest = {"seed": SEED, "panel": "MYP-26", "injected": log}
json.dump(manifest, open("injected_defects.json", "w"), indent=2)
```

The yield panel is planted differently. Its errors are not in the yield column at all; they are in the production and area columns that produce it. A production figure filed in tonnes rather than thousand tonnes multiplies the implied yield by a thousand, which is unmistakable; an area filed in acres rather than hectares divides it by 2.471, which is subtle enough to survive a casual glance. Both are recoverable because the yield is a ratio of two quantities whose plausible range is known for each crop.

2.5 Why this panel is a fair test

Three properties make MYP-26 useful as a test bed. It has a complete target grid, so completeness can be measured against a known denominator. Its defects are of the kinds that occur in the named public sources, not arbitrary corruption. And its true values are generated before the defects are applied, so precision and recall of the cleaning rules can both be computed. A strategy that only reports what it changed, without knowing what it should have changed, cannot distinguish diligence from over-cleaning.

The panel is still hypothetical, and Section 12 and Appendix F are explicit about where real data would differ: irregular publication calendars, PDF-only tables, retrospective revisions that change history, and suppression rules that vary by state. The method is designed for those complications; the test panel is simply the cleanest way to prove the mechanics.

3. Data Quality in Agribusiness Datasets

Agricultural data is not uniquely dirty, but it is dirty in domain-specific ways. A price series is produced by a market that closes, a crop is named differently in three states, a yield estimate is revised twice after its first release, and a district is split into two districts with new codes. None of these is random noise. Each has a cause, and the cause determines the correct treatment.

3.1 Six structural causes

Before listing symptoms, it helps to name the mechanisms that generate them.

- **Unit plurality —** prices are recorded per quintal in one market, per kilogram in another, and per tonne in a trade series. The same physical quantity arrives in three numbers.
- **Nomenclature drift —** a crop has a botanical name, a notified name, a local name and a trade name. 'Paddy(Dhan)', 'Rice' and 'Paddy' may refer to the same commodity at different stages of processing.
- **Administrative change —** districts are renamed, split and merged; states were reorganised in 2014. A name is not a stable identifier, and a join on names silently duplicates or loses history.

- **Revision and vintage —** advance estimates are replaced by second estimates and then revised estimates. The number for 2024 changes after 2024 ends, so a model trained today on '2024' is training on information that did not exist in 2024.
- **Calendar structure —** mandis close for festivals, weekly markets meet on fixed days, and some commodities trade only seasonally. A blank cell may mean 'no trade', not 'data lost'.
- **Suppression and latency —** small counts are suppressed, some series are discontinued, and some states report late. The absence is a publication decision, not a measurement failure.

These causes produce the eight defect classes below. The classes are not a list of annoyances; they are a diagnostic vocabulary. Naming the class correctly is most of the treatment, because each class has a different repair and a different legitimate fallback.

3.2 Eight defect classes

Table 6. *The eight defect classes and the first test that exposes each.*

Class	Typical manifestation	What it breaks	Primary detection
Missing values	Blank, null, dash, 'NA', or absent row	Coverage; model sample; comparability	Null profile by key and period; row-absence check
Duplicates	Re-pulled rows; same key twice with different retrieval times	Counts, sums and any row-weighted statistic	Exact and business-key duplicate counts
Format and type	DD/MM/YYYY vs ISO; prices stored as text with commas	Joins, sorting, arithmetic	Type inference; parse failure rate; regex conformance
Unit and scale	INR/kg vs INR/quintal; tonnes vs '000 tonnes; acres vs ha	Levels and elasticities; any cross-series comparison	Plausibility band against crop-state median
Entity naming	Crop aliases; state and district variants; whitespace	Joins; group-by totals; panel balance	Canonical map; unmatched share; cardinality
Outliers and typos	Decimal slip, digit error, genuine festival spike	Means, variances, model coefficients	Robust z on logs; neighbour corroboration
Inconsistency	Same fact differs across sources; totals don't add	Reconciliation; trust in any single source	Cross-source ratio; sum-of-parts check
Revision and vintage	Advance estimate overwritten by revised estimate	Point-in-time correctness; backtest honesty	Vintage id present; as-of join test

A ninth class is worth naming separately even though it is not a defect in the data: documentation drift. The code that cleans a column and the dictionary that describes it disagree, or the dictionary describes a unit that the source stopped using two years ago. It does not appear in a null count, and it is the most common reason a pipeline that worked last quarter produces a wrong answer this quarter.

3.3 Quality dimensions and thresholds

Quality is not one number. The strategy scores six dimensions, each with an explicit threshold. The thresholds are deliberately demanding for the must-have series and looser for the contextual ones; a single global threshold would either block the build on a minor series or accept a broken critical one.

Table 7. *Six quality dimensions with thresholds. Accuracy is measured, not assumed, and is the only dimension that needs an external benchmark.*

Dimension	Definition	Threshold	Action on failure
Completeness	Share of expected cells with a usable value	â‰¥ 98% for must-have price series; â‰¥ 90% otherwise	Block the series from modelling; investigate source
Validity	Share of values inside the plausible domain	100% after repair, or explicitly flagged	Repair with evidence, else flag and exclude
Uniqueness	Duplicate business keys	0	Resolve before any aggregation

Dimension	Definition	Threshold	Action on failure
Consistency	Agreement across overlapping sources	â‰¤ 5% relative difference, or explained	Reconcile by documented precedence, never average
Timeliness	Lag from reference period to availability	Within the stated publication calendar	Mark provisional; do not backfill silently
Accuracy	Agreement with an independent benchmark	Within the tolerance set per indicator	Downgrade the indicator; record the discrepancy

Completeness is the dimension most often reported and least often defined. Here it has a denominator: the full expected grid for the declared grain. A series that is 100% complete within the rows it happens to contain can still be 60% complete against the grid. The distinction is the difference between a dataset that looks clean and one that is usable.

3.4 How defects become wrong answers

It is tempting to treat cleaning as a cosmetic step before the real work. The four failure modes below show why it is not. Each is a path from a specific defect to a specific kind of wrong conclusion, and each is addressed by a named rule later in the document.

- **Bias from unit and scale —** a 100× unit error in 1% of cells shifts a crop mean by roughly the error times the share. A ₹4,000 quintal price entered as ₹40 per kilogram is not an outlier to be winsorised; it is a unit to be converted. Treating it as noise biases the level downward.
- **Leakage from vintage —** if revised production estimates are used to train a model that is then evaluated on the advance estimates it would have seen, the backtest is optimistic. The model has been shown the future.
- **Aggregation error —** averaging district yields to obtain a state yield gives every district equal weight, so a district with 2% of the area contributes as much as one with 40%. The state figure is then not the state's yield.
- **False precision from over-cleaning —** aggressive filtering of 'outliers' deletes genuine scarcity events, which are exactly the observations a price model needs. The remaining series is well behaved and uninformative.

> **THE ASYMMETRY THAT GOVERNS THE STRATEGY**
>
> A missed defect usually adds noise; an over-aggressive repair adds bias and destroys evidence. Where the two risks conflict, the strategy prefers to flag rather than overwrite, and to publish the flag rather than hide it.

4. Governing Principles

Seven principles decide the many small judgements that a cleaning strategy cannot enumerate in advance. They are stated as constraints rather than aspirations, because a principle that cannot be violated is not useful. Each has a practice that implements it and a test that shows whether it held.

P1 — Raw is immutable

Every pull is written once to a dated snapshot and never edited. All repair happens downstream, so the original evidence survives every later decision.

Test of compliance: A hash mismatch on any raw file fails the build.

P2 — Classify before repairing

A gap is classified by cause — closed market, missing file, suppression, late report — before any method is chosen. The cause selects the treatment; the symptom does not.

Test of compliance: Every null in the clean table carries a non-null quality flag.

P3 — Rules, not edits

Each intervention is a named function with parameters, a rule identifier and a unit test. No value is changed by hand in a notebook.

Test of compliance: The cleaning log contains only rule identifiers from the register.

P4 — Measure the change

Row counts, completeness, duplicate counts and imputation shares are reported before and after every step. A step that changes more than 2% of rows needs a written justification.

Test of compliance: The scorecard is regenerated on every run and compared with the previous one.

P5 — Smallest sufficient intervention

Repair only when the evidence determines the correct value. Otherwise flag, exclude from the relevant use, and leave the value visible.

Test of compliance: Corrections are reversible; flags are not deletions.

P6 — Preserve provenance

Every fact carries its source, vintage, retrieval time and quality flag. A clean value that cannot be traced to a source is not clean; it is unattributed.

Test of compliance: No fact row has a null source, vintage or flag.

P7 — Reproducible by construction

The pipeline is deterministic, versioned and runnable from a clean clone. Reproducibility is a property of the design, not a heroic effort at the end.

Test of compliance: A clean clone plus the sealed seed reproduces the panel hash.

4.1 The principles in tension

The principles conflict in predictable places, and the strategy resolves the conflicts in advance rather than case by case. Completeness pushes toward imputation; smallest intervention pushes against it. Reproducibility pushes toward freezing rules; accuracy pushes toward revising them when a source changes. The resolutions are:

Table 8. *Five recurring conflicts and their standing resolutions.*

Tension	Resolution	Recorded where
Impute vs leave missing	Impute only short internal gaps where neighbouring values bound the answer; otherwise leave missing and flag.	R08; §6.1
Correct vs flag an outlier	Correct only when a mechanical cause (unit, decimal place) determines the value; otherwise flag and retain.	R06–R07; §6.2
Reconcile vs choose a source	Use a documented precedence order; never average disagreeing sources into a number that neither published.	R12; §6.4
Freeze vs update a rule	Rules are versioned; a change requires a new version, a re-run and a diff of the scorecard.	§9.3
Aggregate vs preserve detail	Aggregate with a ratio of sums, and keep the detail table so the aggregate can be audited.	R09; §7.3

Writing these down matters more than it might appear. In practice, the costly arguments about data quality happen late, when a result looks wrong and no one remembers why a particular cell was changed. A standing resolution turns that argument into a lookup: the rule identifier in the log points to the decision, and the decision points to the principle.

5. Step-by-Step Cleaning Plan

The plan has ten steps, S0 to S9. They run in a fixed order because the output of each is the precondition of the next: a duplicate cannot be resolved until its key is canonical, a unit error cannot be detected until the entity is identified, and an outlier cannot be judged until the unit is right. Reordering the steps is not a stylistic choice; it changes which values are flagged.

Each step below states its purpose, its inputs, the actions it takes, the decision rule that governs ambiguous cases, its output and its gate. The rule identifiers refer to the register in Appendix C, which also carries the parameter values and the test for each rule.

5.1 The ten steps at a glance

Table 9. *The ten cleaning steps. The four gates are the same gates shown in the workflow diagram.*

Step	Name	Core action	Rules	Gate
S0	Intake and freeze	Write an immutable snapshot, capture provenance and hash	—	Snapshot hash recorded
S1	Profile and score	Profile every column and score six quality dimensions	—	G1: defect register complete
S2	Schema and domain validation	Enforce types, ranges, enums and cross-field logic	R01	G2: zero unexplained type failures
S3	Deduplicate	Resolve exact and business-key duplicates with a survivorship rule	R02	G2: zero duplicate keys
S4	Harmonise entities and units	Map crops, geographies, mandis, units and currency	R03, R06, R10	G3: zero unmapped entities
S5	Dates and calendars	Parse to ISO, derive crop/fiscal year, attach trading calendar	R04, R05	G3: zero unparsed dates
S6	Missing values	Classify every gap by cause and apply the bounded treatment	R08	Completeness â‰¥ 98%
S7	Outliers	Flag robust outliers and separate errors from genuine events	R07, R14	G4: every flag has a reason
S8	Reconcile sources and vintages	Apply precedence, keep every release, build as-of views	R11, R12, R13	G4: point-in-time test passes
S9	Validate and freeze	Run the suite, publish the scorecard, sign off and hash	R14	G4: clean layer frozen

On MYP-26 the plan fires 550 logged interventions across 9 rules. The distribution is informative: most interventions are harmonisation and gap treatment, not correction of values. That is the normal shape of cleaning a real panel. Corrections are rare and consequential; harmonisation is frequent and mechanical; imputation sits in between.

5.2 S0 — Intake and freeze

The first step is to make the raw data impossible to lose and impossible to alter. Each pull is written to a dated file under an immutable path, and a manifest row records the source, the endpoint, the retrieval timestamp, the record count, the licence and the SHA-256 hash. The raw file is never opened in edit mode again.

- **Manifest granularity —** one row per pull, not one row per source. A monthly series pulled twelve times has twelve manifest rows, because the twelve snapshots are different evidence.

- **Failures are data —** the manifest is append-only. A failed or partial pull is recorded as failed, not deleted, so the coverage gap can be explained later.
- **Capture the context —** the source's own identifier, title, unit label and footnote are captured beside the data. They are often the only clue to a unit change.

Output: a raw folder containing dated files and a manifest. Gate: every file in the folder has a manifest row and a hash that matches on re-computation.

5.3 S1 — Profile and score

Profiling answers three questions for every column: what is in it, what is missing from it, and what cannot be parsed. The profile is generated, not inspected by eye, because the interesting failures are usually in the tail of a distribution or the edge of a period range.

- **Column profile —** row count, distinct keys, period range, null share, zero share, minimum, maximum, median and the five most common values, per column and per series.
- **Row-absence check —** the expected grid is generated from the scope (crops × states × months), then left-joined to the data. Rows that do not exist are counted as missing. This single check catches more real problems than every other profile combined.
- **Defect register —** each defect is entered in the defect register with a class, a severity, an example row and the rule expected to resolve it. The register is the work queue for S2–S8.

Baseline scorecard for MYP-26: completeness 92.3%, exact duplicate rate 2.2%, non-ISO dates 3.3%, alias share 1.6%. These are the numbers the later steps must improve, and the numbers the final report quotes.

Output: a profiling report and a prioritised defect register. Gate G1: every column has a profile, every expected series has a coverage figure, and every critical defect has an owner and a rule.

5.4 S2 — Schema and domain validation

Validation is where the pipeline refuses to guess. A price stored as text with a thousands separator is coerced to a number and logged; a price stored as 'not available' is nulled and logged; a price of â^'40 is rejected and logged. Silent coercion is forbidden because it converts a visible source problem into an invisible analytical one.

Table 10. *Validation checks and the response to failure. The default response is to null and log, not to coerce.*

Check	Rule	On failure
Type	Numeric field parses as a number	Coerce if unambiguous; else null and log
Domain	Price > 0, production â‰¥ 0, area > 0, share in [0, 1]	Null invalid values; never clip silently
Enum	Crop, state, unit, flag drawn from a controlled list	Reject the row and route to the mapping queue
Cross-field	Production and area imply a plausible yield	Flag for S4 unit repair
Referential	Every source_id exists in the source register	Block the load; a missing source is a governance failure
Key	Business key is unique after harmonisation	Route to S3; never aggregate a duplicated key

The domain check is not cosmetic. A negative price is impossible, and leaving it in the data would distort every mean and every log transform downstream. A zero price is possible but means something different from a price of zero rupees: it means no trade. S2 distinguishes the two, because S6 will treat them differently.

5.5 S3 — Deduplicate

There are three kinds of duplicate, and conflating them is a common source of error.

- **Exact —** byte-identical rows from a repeated load. Safe to drop after keeping the latest retrieval.

- **Key —** the same business key with different values, usually because the source was re-pulled or revised. Requires a survivorship decision, not a blind drop.
- **Near —** the same entity with slightly different names or numbers, often after harmonisation. Requires fuzzy matching and manual review.

The survivorship rule for MYP-26 is stated once and applied everywhere: prefer the row that carries a usable price, then the most recent retrieval, then the row with the most populated fields. Ties are broken by source precedence (Appendix C). The rule is deterministic, which means the same input always produces the same survivor.

In the reference run, 79 duplicate rows were dropped on the business key, taking the duplicate rate from 2.2% to zero. Every drop is in the log with the survivor's row identifier, so a reviewer can reconstruct the decision. Duplicates are never summed: summing a re-pulled price doubles the weight of that month in any aggregate, which is a silent bias rather than a visible error.

5.6 S4 — Harmonise entities and units

Harmonisation is the largest step by row count and the one that most affects joins. It has four masters, each with one job.

Table 11. *The four reference masters. Codes, not names, are the join keys.*

Master	Key	Contents	Failure it prevents
Crop master	crop_code	Canonical name, synonyms, botanical name, variety flag	Paddy/Rice split into two crops
Geography master	LGD code	State and district codes with effective dates	Renamed or split district losing history
Mandi register	mandi_id	Market name, district, state, active dates	Two markets with one name
Unit table	unit_code	Conversion factor to the canonical unit	INR/kg and INR/quintal in one column

Unit repair is the part that requires domain knowledge rather than a lookup. The rule is to compare each value with a robust median for its crop–state series and act only when the ratio is diagnostic. A ratio near 0.01 is a per-kilogram entry; a ratio near 10 or 0.1 is a one-place decimal slip. Anything else is left alone and flagged for review.

Table 12. *Unit-repair decision table. The bands are wide enough to absorb seasonality and tight enough to distinguish a unit error from a bad year.*

Observed ratio to series median	Interpretation	Action	Rule
0.004 – 0.025	Price in INR/kg under a per-quintal label	Multiply by 100	R06
0.05 – 0.20	One decimal place low	Multiply by 10	R06
5 – 20	One decimal place high	Divide by 10	R06
0.20 – 5 (outside robust band)	Genuine variation or an unresolved error	Flag; do not alter	R07/R14
Yield > 8× crop median	Production in tonnes under a '000-tonne label	Divide production by 1,000	R10
Yield < 0.55× crop median	Area in acres under a hectare label	Divide area by 2.471	R10

The reference run corrected 32 unit-drift cells and 30 decimal slips, and repaired 6 production and 4 area records in the yield panel. The important number is not the count but the ordering: unit repair happens before outlier flagging, so the outlier screen sees prices in one unit. Reversing the order would flag every per-kilogram entry as an extreme low value and every corrected value as an extreme high one.

5.7 S5 — Dates and calendars

A date is an identifier, not a string. The parser accepts the formats present in the source (ISO, DD/MM/YYYY, Mon-YYYY), resolves ambiguity by source convention rather than by guessing, and writes an ISO month start. Where day and month are both â‰¤ 12 and the source convention is unknown, the row is quarantined rather than guessed: a silently swapped date moves a price by a month and is almost impossible to detect downstream.

- **Three calendars —** ISO 8601 period start, plus explicit crop_year, fiscal_year and marketing_year columns. Indian fiscal years and crop years do not align with the calendar, and a model that ignores the difference mis-specifies the season.
- **Trading calendar —** a state-specific trading calendar marks Sundays, festivals and weekly closures. It is used to explain zeros, never to forward-fill prices for modelling.
- **Period vs vintage —** each row keeps retrieved_at and vintage_id alongside the period. The period says what the row describes; the vintage says when it became knowable.

5.8 S6 — Missing values

Missing values are treated as a classification problem before they are treated as an imputation problem. The taxonomy below is the core of the method; the imputation rule is deliberately narrow.

Table 13. *The missing-value taxonomy. Only one of the six classes is eligible for imputation.*

Class	Cause	Example	Treatment
Structural	Market closed, no trade, seasonal absence	Sunday; onion not traded in June	Keep missing; attach calendar feature; never impute
Suppressed	Publisher withholds a small count	District yield withheld	Keep missing; flag; never back-calculate
Not reported	Source missed a period	State did not file for a month	Impute only if internal gap â‰¤ 3 months; else leave
Lost	File or field missing in the pull	Column absent from one release	Re-pull; if impossible, treat as not reported and log
Not applicable	Concept does not apply	Variety price where no variety market exists	Keep null; exclude from that indicator
Unknown	Cause not determinable	Blank with no footnote	Leave missing; sensitivity analysis; do not impute

The imputation rule is bounded in three ways: by gap length, by neighbours and by the plausible range. An internal gap of up to three months is filled by time interpolation between the surrounding observed values. A longer gap is left missing, because interpolation across a season invents a shape the data does not support. Edge gaps are never filled, because there is no second anchor.

Table 14. *Gap treatment by length and position. The three-month limit is a parameter, not a universal law; Appendix F explains how to reset it for weekly or daily data.*

Gap	Length	Method	Justification
Internal, short	1–3 months	Linear interpolation in time	Neighbours bound the value; seasonal shape is preserved locally
Internal, long	> 3 months	Leave missing; flag	Interpolation would cross a season and smooth away the signal
Leading or trailing	Any	Leave missing; flag	One-sided anchor; extrapolation is not imputation

Gap	Length	Method	Justification
Structural	Any	Leave missing; calendar feature	The absence is the fact
Target variable	Any	Never impute	Imputing the target trains the model on invented outcomes

On MYP-26, 156 cells were imputed (4.5% of the panel), all from internal gaps of three months or less. 41 zero prices and 6 invalid values were classified as structural or invalid and left missing. No long gap was interpolated. The imputation share is published because a clean table with a high imputation share is a partly synthetic table, and the reader is entitled to know that before trusting a forecast.

> **WHY NOT MEAN-IMPUTE AND MOVE ON**
>
> Mean imputation preserves the sample mean and destroys the variance, the covariance with every other variable, and the time structure. In a price panel it also erases seasonality by replacing a December onion price with an annual average. It is fast, and it is the single most common way a clean-looking panel becomes an uninformative one.

5.9 S7 — Outliers

Outlier detection runs on the log of the price, within each crop–state series, using a modified z-score built on the median absolute deviation. The log makes a doubling and a halving symmetric; the median and MAD make the threshold resistant to the very values being tested. A conventional mean and standard deviation would be pulled toward a large error and could fail to flag it.

```
log_price = np.log(price[price > 0])
med  = np.median(log_price)
mad  = np.median(np.abs(log_price - med))
scale = 1.4826 * mad
z = 0.6745 * (log_price - med) / scale
flag = abs(z) > 3.0
```

The threshold of 3.0 is a screening parameter, not a verdict. Three outcomes follow.

- **Mechanical error — correct —** a ratio to the series median near a power of ten, or a ratio near 0.01, identifies a mechanical cause. The value is corrected by the unit rule and the correction is logged. This is the only case in which an outlier is overwritten.
- **Genuine event — retain and flag —** a large deviation that is corroborated by neighbouring months, by another market, or by a documented event is retained and flagged as a genuine spike. Deleting it would delete the scarcity signal the model is meant to learn.
- **Unexplained — flag and exclude —** a large deviation with no corroboration and no mechanical cause is flagged and excluded from model training, but left visible in the clean table. It is not winsorised by default, because replacing it with a percentile invents a value and hides the uncertainty.

The reference run flagged 1 genuine event and 3 unexplained values, and rejected 2 values outside the plausible band. Five of the seven planted digit errors were therefore caught — three as unexplained outliers and two as implausible — and two stayed inside the robust band. That residual is not a bug to hide: it is the measured limit of a univariate screen on a seasonal series, and §6.2 extends the screen with a seasonal-residual test for exactly this reason.

Output: a clean price table where every value is either observed, mechanically corrected, imputed under the bound, or flagged. Gate: every flag has a rule, a reason and a review status; no value is both flagged and silently used in training.

5.10 S8 — Reconcile sources and vintages

Two sources rarely agree exactly, and a third usually disagrees with both. The strategy does not average disagreement away. It applies a documented precedence order, records the difference, and investigates anything above a tolerance.

- **Precedence by indicator —** official statistics over trade estimates for production; the designated price series over a secondary aggregator for price. Precedence is set per indicator in the source register, not chosen per row.
- **Tolerance —** a relative difference above 5% between overlapping sources is investigated and explained. Below 5%, the higher-precedence source is used and the difference is recorded. Above 5%, the series is flagged until the cause is known.
- **Point-in-time —** every release is kept with its vintage_id. A fact is retrieved as of the vintage that was available at the forecast origin, using an as-of join. The latest value is never substituted into a historical row.
- **Suppression —** suppressed values are retained as null and flagged. They are not inferred from neighbouring cells, because the suppression exists to prevent exactly that inference.

The yield panel carries 10 release identifiers across 2020–2025, with 2024 and 2025 having three releases each. The point-in-time view selects 288 rows — one per crop, state and year — using the vintage available at the time. Training on the latest revised series and backtesting against it would be a look-ahead error even though every individual number is correct.

5.11 S9 — Validate and freeze

The final step is a gate, not a formality. The suite is run, the scorecard is regenerated, the outputs are hashed, and the clean layer is frozen. If any gate fails, the scope is narrowed or the series is dropped; the build does not proceed on a failed gate.

Table 15. *The seven-check acceptance suite. All checks pass on the reference run.*

Check	Acceptance	Measured on MYP-26
Completeness	â‰¥ 98% of must-have price cells	98.5%
Duplicate keys	0	0
Date conformance	100% ISO	100%
Entity conformance	100% mapped to a master	100%
Imputation share	< 5% of price cells	4.5%
Yield derivability	100% of yield rows recomputable from production and area	100%
Determinism	Clean clone reproduces the panel hash	verified in §9.5

Output: a frozen clean layer, a cleaning log, a scorecard and an updated data dictionary. Gate G4: the suite passes, the log is complete, and a second run from the raw snapshot produces an identical hash.

One deliberate omission: there is no step for 'manual correction'. If a value needs a human decision, the decision is captured as a parameter of a rule or as an entry in a review queue, not typed into the table. That is what keeps the process reproducible when the person who made the decision is no longer on the project.

6. Method Playbooks

Section 5 says what happens and in what order. This section says why each method was chosen, what it preserves, what it costs, and when it should be replaced. The playbooks are written to be argued with: each states the conditions under which a different method would be correct.

6.1 Missing data

The first question is not 'which imputation method' but 'why is the value missing'. In agricultural panels the mechanism is often observable, which makes classification more reliable than in clinical or survey data.

6.1.1 Mechanisms in agricultural terms

Table 16. *Missingness mechanisms and what each permits. MNAR is common in price reporting and is the reason flags are mandatory.*

Mechanism	Meaning	Agricultural example	Consequence
MCAR	Missingness independent of everything observed or unobserved	A file failed to transfer for a random month	Complete-case analysis is unbiased, though wasteful
MAR	Missingness explained by observed variables	Small states report less often; reporting depends on state and crop	Conditional imputation can be unbiased if the model is right
MNAR	Missingness depends on the unobserved value	High prices discourage reporting; only low prices are filed	No imputation from the observed data is unbiased; flag and sensitivity-test

The strategy classifies by cause where the cause is recorded and treats unexplained gaps as MNAR-until-shown-otherwise. That is conservative on purpose. A model that imputes an unexplained gap at the series mean assumes MAR without evidence, and in a price series the assumption is frequently wrong: markets with missing quotes are often the thin markets whose prices differ most.

6.1.2 Seven methods compared

Table 17. *Imputation methods. The chosen method for MYP-26 is interpolation, bounded to short internal gaps; the rest are fallbacks with stated preconditions.*

Method	Preserves	Destroys / risks	Cost	Use when
Listwise deletion	Nothing artificial; simple	Power; biases if not MCAR; breaks panel balance	Low	Gap is long, unexplained, and the series is not critical
Median / mean	Central tendency	Variance, covariance, seasonality, time structure	Low	Almost never for a time series; acceptable for a static covariate with <2% missing
Last observation carried forward	Level at the last observation	Trend and turning points; creates flat runs	Low	Only for a stock variable known not to change between observations
Linear interpolation	Local level and slope between anchors	Curvature; invents values across long gaps	Low	Short internal gaps (â‰¤3 months) in a smooth series
Seasonal-naive fill	Seasonal shape	Level drift; copies last year's shock	Low	Strongly seasonal series with a stable year-on-year pattern
KNN / MICE	Cross-variable and temporal structure	Runtime; model dependence; over-smoothing; hard to audit	High	Many correlated indicators and a gap pattern that is demonstrably MAR
Model-based (state space)	Trend, season, uncertainty	Specification risk; complexity	High	Target series where an interval is needed, not just a point

Interpolation is chosen for the reference panel because the gaps are short, the series are smooth at monthly frequency, and the method is transparent: a reviewer can see the two anchors and the straight line between them. MICE would produce a more sophisticated number with no way to audit it, and on a gap of one or two months it would add complexity without adding information.

There is one case where interpolation is clearly wrong: a price series with a structural break inside the gap, such as a policy announcement or a market closure that lasted the whole gap. The rule cannot know about the announcement, which is why the imputed cells are flagged and the sensitivity analysis in §6.1.4 is run on every indicator that uses them.

6.1.3 Bounds and guardrails

An imputed value is a claim about an unobserved quantity, and the claim should be bounded by what is physically possible. Every imputed price is checked against four bounds before it is written:

- **Anchor bound —** the value lies between the two observed anchors. Interpolation satisfies this by construction; a model-based method may not.
- **Domain bound —** the value lies inside the crop–state plausible band used by R14. A price of ₹40,000 per quintal for wheat is not made more credible by being imputed.
- **Movement bound —** the imputed value does not move the series more than 25% from the mean of its anchors. A larger move implies the gap spans a regime change, and the gap should be left missing.
- **Eligibility bound —** no imputation is applied to the target variable, to a suppressed cell, or to a structural absence. These are exclusions, not bounds.

On MYP-26 the guardrails bind on 156 cells, 4.5% of the panel. All are internal gaps of three months or less; no long gap and no edge gap was filled. 41 zero prices were classified as structural and left missing, which is why the completeness figure after cleaning is 98.5% rather than 100%: the remaining 1.5% is information, not failure.

6.1.4 Sensitivity analysis

The effect of imputation is measured, not assumed. The pipeline produces three versions of every indicator that uses an imputed cell: the primary series, a series with the imputed cells dropped, and a series with the imputed cells replaced by the series median. If a headline result changes sign or significance between the three, the result is reported as imputation-dependent.

Table 18. *Three-way sensitivity check for any indicator that consumes imputed values.*

Version	Construction	What a difference proves
Primary	Imputed values included, flagged	The result as reported
Drop-imputed	Imputed cells set to missing; indicator computed on observed only	Whether the result depends on invented values
Median-fill	Imputed cells replaced by the series median	Whether the result depends on the imputation method rather than the data

> **A RULE OF THUMB THAT SURVIVES CONTACT WITH REAL DATA**
>
> If more than 5% of a series is imputed, do not report a point forecast from it without an interval that includes the imputation uncertainty. If more than 10% is imputed, drop the series from the model and report the coverage gap instead.

6.2 Outliers and transcription errors

An outlier is a value that is surprising given a model of how the data is generated. That definition makes the model explicit, which is useful: a value can be an outlier under one model and perfectly ordinary under another. The strategy uses three screens in increasing order of context, and treats their outputs differently.

6.2.1 Three screens

Table 19. *Three outlier screens. No single screen is trusted; a flag is confirmed or dismissed by the next.*

Screen	Statistic	Strength	Weakness	Role
Univariate robust	Modified z on log price; MAD scale	Immune to the values being tested; simple	Ignores seasonality and cross-market information	First pass; catches decimal slips and unit errors
Distributional	IQR fences (1.5× and 3×)	Familiar; no distributional assumption	Sensitive to a strongly seasonal series; arbitrary fences	Cross-check, not a decision rule
Contextual / residual	STL or seasonal-naive residual; neighbour corroboration	Separates a seasonal peak from an error; uses time structure	Needs a stable seasonal pattern and enough history	Second pass; resolves ambiguous flags

The contextual screen is what distinguishes this strategy from a generic z-score filter. A festival spike and a decimal slip can have the same univariate z. The spike is corroborated by adjacent months and by other markets; the slip is not. The screen below formalises the comparison.

```
# contextual corroboration for a flagged value
neighbour_median = median(series[t-1], series[t+1])       # same crop-state
cross_market     = median(price[other states, crop, t])
ratio_local      = value / neighbour_median
ratio_cross      = value / cross_market

if 0.7 <= ratio_local <= 1.4 and 0.7 <= ratio_cross <= 1.4:
    flag = 'spike_genuine'        # retain; it is a real market event
elif is_mechanical(value, series_median):
    flag = 'corrected'            # unit or decimal place determines the value
else:
    flag = 'outlier_unresolved'   # retain visibly; exclude from training
```

6.2.2 Typo or event? A decision matrix

Table 20. *Decision matrix for an anomalous value. The default when evidence is absent is to flag, not to delete.*

Evidence	Mechanical error	Genuine event	Decision
Ratio to series median near 0.01, 0.1 or 10	Yes	Unlikely	Correct by factor; log
Corroborated by adjacent month	No	Yes	Retain; flag genuine
Corroborated by other markets	No	Yes	Retain; flag genuine
Matches a documented event (festival, drought, policy)	No	Yes	Retain; annotate the event
Isolated, no corroboration, no mechanical factor	Maybe	Unlikely	Flag unresolved; exclude from training
Outside the physical domain	Yes	No	Correct if factor is diagnostic, else reject

In the reference run R06 had already corrected the decimal slips and unit-drift cells before the outlier screen ran, so the screen inherited a series in one unit and at one scale. A genuine spike was still flagged and retained, and 3 values were flagged unresolved. Two of the seven planted digit errors stayed inside the robust band because the seasonal variation in their series was large enough to inflate the MAD scale. That is the honest limit of a univariate screen, and it is why the contextual screen is specified as the second pass rather than as an optional refinement.

6.2.3 Treatment policy

- **Correct —** correct only when the cause determines the value. A unit error and a one-place decimal slip are determinate; a digit change from 4,000 to 4,800 is not.
- **Retain —** retain and flag values that are surprising but corroborated. The flag travels with the value into every downstream table.

- **Exclude —** flag and exclude uncorroborated values from model training, while leaving them visible in the clean table. Exclusion is a modelling decision; deletion is a data decision, and they are not the same.
- **Do not —** never winsorise a price by default. Winsorising replaces the most informative observations with a percentile and makes the series look better behaved than it was.

6.3 Entity and unit harmonisation

6.3.1 Crop master and fuzzy matching

The crop master holds a canonical code, the notified name, common synonyms and a variety flag. Matching is exact on a normalised string first (case-folded, whitespace-collapsed, punctuation removed), then fuzzy with a similarity threshold, then manual. The thresholds matter because a false merge is worse than a missed merge: merging 'Rice' and 'Paddy' silently changes what is being priced, while failing to merge them produces an obvious duplicate series that someone will notice.

Table 21. *Fuzzy-matching thresholds. The asymmetric cost of a false merge sets the conservative cut-off.*

Similarity	Action	Example
1.00 after normalisation	Automatic map	'SOYABEAN' â†’ SOYBEAN
0.92 – 0.99	Automatic map with log entry	'Wheat FAQ' → WHEAT, variety=FAQ
0.80 – 0.91	Manual review queue	'Onion (Nashik)' → ONION, market attribute
< 0.80	No match; reject the row	'Bajra' is a different crop, not a typo

6.3.2 Geography with effective dates

Names change; codes persist. Every geography row carries an LGD code and a validity interval. A district split in 2018 has one parent code valid to the split date and two child codes valid after it. Aggregating across the split requires an explicit mapping decision, not a string replace.

Table 22. *Geography resolutions. The code is the key; the name is a label with a validity period.*

Raw name	Period	Canonical resolution	Method
Telangana (new state)	2014 onward	LGD 36, carved from Andhra Pradesh	Effective-dated map
Maharastra	All	Maharashtra, LGD 27	Spelling map
MP	All	Madhya Pradesh, LGD 23	Abbreviation map
Ahmednagar	2018 onward	Ahilyanagar, same LGD code	Rename; code unchanged
Pune (rural)	All	Pune district, with a market attribute	Name is not a district key

6.3.3 Units and currency

Table 23. *Conversion table. Factors live in code and in the unit table, never in a comment.*

Quantity	Common source units	Canonical	Conversion
Price	INR/kg, INR/quintal, INR/tonne	INR/quintal	1 quintal = 100 kg = 0.1 tonne
Production	tonnes, '000 tonnes, lakh tonnes	'000 tonnes	1 lakh = 100; 1 tonne = 0.001 '000 t
Area	acres, hectares, '000 hectares	hectares	1 ha = 2.471 acres
Yield	kg/ha, t/ha, quintal/ha	t/ha	1 t/ha = 10 quintal/ha = 1,000 kg/ha
Currency	USD, EUR, INR	INR	Dated RBI reference rate; store the rate with the row
Deflation	Nominal INR	Real INR (2011-12 base)	CPI or WPI, with the index vintage recorded

Currency conversion has a subtlety that unit conversion does not: the rate is dated. A USD price converted at today's rate and joined to a 2019 INR series mixes a 2019 quantity with a 2026 relative price. The rate must be the one in force at the observation date, and it must be stored with the row so the conversion is reversible.

6.4 Duplicates, conflicts and survivorship

Survivorship is the rule that decides which row lives when two rows describe the same fact. It is a policy, not a data property, and it must be written before the first conflict is encountered. The MYP-26 order is:

- **1 —** source precedence: the designated series for that indicator wins over a secondary aggregator.
- **2 —** vintage: the release that was current at the retrieval date wins over an older snapshot, unless the analysis is explicitly point-in-time.
- **3 —** completeness: the row with more non-null fields wins.
- **4 —** recency: the later retrieval wins.
- **5 —** stable tie-break: the lower row identifier wins, so the outcome is deterministic.

Conflicts across sources are not resolved by averaging. An average of two disagreeing sources is a number that neither source published, and it hides the disagreement that a reader needs to see. The strategy records both values, uses the precedence winner, and reports the difference in the reconciliation table.

Table 24. *Source precedence by indicator. The tolerance is the point at which a difference stops being rounding.*

Indicator	Precedence 1	Precedence 2	Tolerance	On breach
Production	DES	FAOSTAT	5%	Investigate; report both; do not average
Wholesale price	Agmarknet	data.gov.in mirror	3%	Use Agmarknet; record difference
Yield	DERIVED (production ÷ area)	DES reported yield	5%	Prefer derived; log discrepancy
Trade	UN Comtrade	ITC Trade Map	5%	Investigate HS-code and period alignment
Exchange rate	RBI reference rate	World Bank	1%	Use RBI; store both

6.5 Revisions, vintages and suppression

Official agricultural estimates are revised, sometimes more than once. An advance estimate for 2024 published in July 2024 is not the same fact as the revised estimate published in February 2026, even though both are labelled '2024'. A dataset that keeps only the latest value has thrown away the information needed to backtest honestly.

Table 25. *A revision triangle in miniature. A point-in-time join selects only the releases knowable at the forecast origin.*

Release	Published	Production (illustrative)	Available for a July 2024 forecast?
Advance	2024-07-14	1,480 kt	Yes, at the release date
Second	2024-10-01	1,450 kt	No
Revised	2026-02-15	1,502 kt	No

The as-of join is the mechanism: for each forecast origin, select the latest release whose publication date is on or before the origin. This is a single SQL predicate, and it is the difference between a backtest that means something and one that flatters the model.

```
SELECT f.*
FROM   fact_yield f
JOIN   forecast_origin o
       ON f.crop_code = o.crop_code
      AND f.state_lgd = o.state_lgd
      AND f.crop_year = o.crop_year
WHERE  f.available_at <= o.origin_date
QUALIFY ROW_NUMBER() OVER (
        PARTITION BY f.crop_code, f.state_lgd, f.crop_year
```

```
        ORDER BY f.available_at DESC) = 1;
```

Suppression is handled by doing nothing to the value and everything to the metadata. A withheld cell stays null, is flagged suppressed, and is excluded from any total that would imply a value. It is not inferred from a published total, because the arithmetic difference is exactly what the suppression was designed to prevent.

The same discipline applies to discontinued series. A series that stops in 2022 is not missing at random after 2022; it has ended. It is flagged discontinued, retained for history, and excluded from the current-period feature set. Imputing a discontinued series forward is the most confident way to be wrong.

7. Preprocessing Framework

Cleaning produces a table that is correct about the past. Preprocessing produces a matrix that is usable for prediction, and the two are not the same artefact. The framework below separates them into layers with explicit contracts, so that a transformation cannot leak future information into a training row without a test failing.

7.1 Five layers

Table 26. *The five layers. Each layer has one job and one gate; a layer never reaches back to repair an earlier one.*

Layer	Contents	Contract	Gate
L0 Raw	Immutable snapshots and manifest	Byte-identical to retrieval; hash recorded	Manifest hash matches
L1 Validated	Typed, parsed, schema-conformant rows	Every value has a type and a domain check	Zero unexplained parse failures
L2 Clean	Harmonised, deduplicated, repaired, flagged	Every intervention logged; every null classified	Quality scorecard passes
L3 Panel	Analysis grain with derived indicators and vintages	One row per crop-state-period; as-of joins only	No feature timestamp after its origin
L4 Matrix	Encoded, scaled, split-ready numeric matrix	Fit on training folds only; schema fixed	Leakage and contract tests pass

The layering is not architectural decoration. It fixes where a decision is allowed to happen. A unit conversion belongs in L2 and must never appear in an L4 feature transform, because an L4 transform is fitted on training data and a unit is a fact about the world. Conversely, a scaler belongs in L4 and must never be applied in L2, because it would be fitted once on the whole panel and leak the test period into training.

7.2 Framework components

Table 27. *Framework components. Each has an artefact that can be reviewed without reading the pipeline code.*

Component	Responsibility	Artefact
Configuration	Declares grain, units, canonical names, thresholds and rule versions	config.yaml, validated on load
Rule registry	Maps every rule identifier to a function, parameters and a test	rules.py + test_rules.py
Transform registry	Declares each derived indicator with its formula, inputs and availability rule	indicators.yaml
Quality monitor	Computes the scorecard before and after each run	scorecard.json + trend table
Provenance store	Records source, vintage and retrieval time for every fact	Columns on every table, not a side file
Contract tests	Asserts schema, dtypes, ranges, uniqueness and leakage rules	test_contract.py, run in CI

The transform registry is the part that pays for itself. Each indicator is declared once, with its formula and the rule that determines when it is available. The feature builder reads the registry rather than hard-coding columns, so a new indicator cannot be added without also declaring its availability. That single constraint prevents most leakage.

7.3 Transformation decisions by variable type

Table 28. *Transformation decisions. The choice is stated with the reason, so a later reviewer can challenge the reason rather than reverse-engineer it.*

Variable	Options	Chosen for MYP-26	Reason
Price level	Level, log, real, differenced, seasonal difference	Real log price, plus 12-month log difference	Stabilises variance; separates inflation from relative price; stationarity

Variable	Options	Chosen for MYP-26	Reason
Yield	Level, log, trend deviation, growth	Level and deviation from a five-year local trend	Annual series is short; differencing throws away the level information
Production / area	Level, log, share, growth	Log level for production; share of state area	Multiplicative scale; share is comparable across states
Weather	Raw, anomaly, z-score, GDD, SPI	Anomaly from a 30-year local normal, plus GDD	A 32 °C day means different things in Punjab and Karnataka
Categorical	One-hot, ordinal, target encoding	One-hot for ≤10 levels; target encoding inside folds above that	Cardinality control without leakage
Counts	Raw, log1p, rate	log1p	Heavy right tail; zeros retained
Calendar	Month dummies, Fourier terms, holiday flags	Fourier terms plus festival and harvest flags	Smooth seasonality with explicit event effects

Two decisions are worth defending. First, price is modelled in real terms with the CPI deflator, because nominal prices in India contain a trend that a model will happily learn as if it were a market signal. Second, yield is not differenced: with six annual observations per series, a difference consumes a third of the history and amplifies measurement error. A deviation from a local trend is the more efficient way to remove the trend without losing the level.

7.4 Scaling, encoding and imbalance

Scaling is model-dependent, and applying it globally is the most common leakage bug in applied machine learning. The rules are simple and non-negotiable:

- **Fit inside the fold —** fit the scaler inside each cross-validation fold, on the training part only. A scaler fitted on the full panel has seen the test period's mean and variance.
- **Scale only where it matters —** tree-based models do not need scaling; regularised linear models, kNN and neural networks do. Scaling a tree model is harmless but adds a step that can leak.
- **Robust where outliers remain —** use a robust scaler (median and IQR) when flagged outliers are retained, because a standard scaler is dominated by the values the strategy chose not to delete.
- **Encode inside the fold —** one-hot encode low-cardinality categories; target-encode high-cardinality ones inside the training fold, with smoothing. Never encode on the full panel.
- **Leave the target alone —** do not scale the target. A scaled target makes the error metric uninterpretable in the units the decision-maker uses.

Class imbalance is a secondary concern for this project because the targets are continuous. It matters for the event indicators — drought, policy intervention, price spike — where the positive class is rare. The approach is to report precision, recall and the confusion matrix rather than accuracy, and to set the decision threshold from the cost of the two errors, not from 0.5.

7.5 Leakage discipline

Leakage is not a single mistake; it is a family of them. The framework tests for five specific forms on every build.

Table 29. *Five leakage forms and the assertion that blocks each. The tests run in CI, not in a notebook.*

Leakage form	How it happens	Assertion
Temporal	A feature uses a value published after the forecast origin	max(available_at) <= origin_date for every feature row
Revision	The latest revised value is used in a historical row	As-of join on available_at; no un-vintaged reads
Transform	A scaler, encoder or imputer is fitted on the full panel	Fit only on the training fold; transform the rest

Leakage form	How it happens	Assertion
Target	A feature is mechanically derived from the target	Registry review; correlation and formula audit
Split	Random split of a time series	Split by origin; rolling-origin evaluation only

7.6 Output contract

The matrix handed to the model is a deliverable with a contract. It declares the grain, the row count, the column list with dtypes and units, the vintage rule, the missing-value policy and the hash of the clean layer it was built from. A model that receives a matrix without this contract cannot be reproduced, and a result from it cannot be defended.

```
grain:        crop_code x state_lgd x period
rows:         3,456
target:       log_real_modal_price, horizon h in {1, 3, 6, 12}
features:     30 declared in indicators.yaml, each with available_at
missing:      imputed_short_gap allowed; structural and long gaps excluded
scaling:      robust, fitted inside each rolling-origin fold
source_hash:  c0a070b878b301f5...
dictionary:   v1.0 (Appendix B and the project data dictionary)
```

8. Workflow and Sequence

The diagram below is the whole strategy on one page. The cleaning lane is loss-aware and fully logged; the preprocessing lane is contract-driven. Seven gates sit between the stages, and the three controls below the lanes apply to every stage rather than to one.

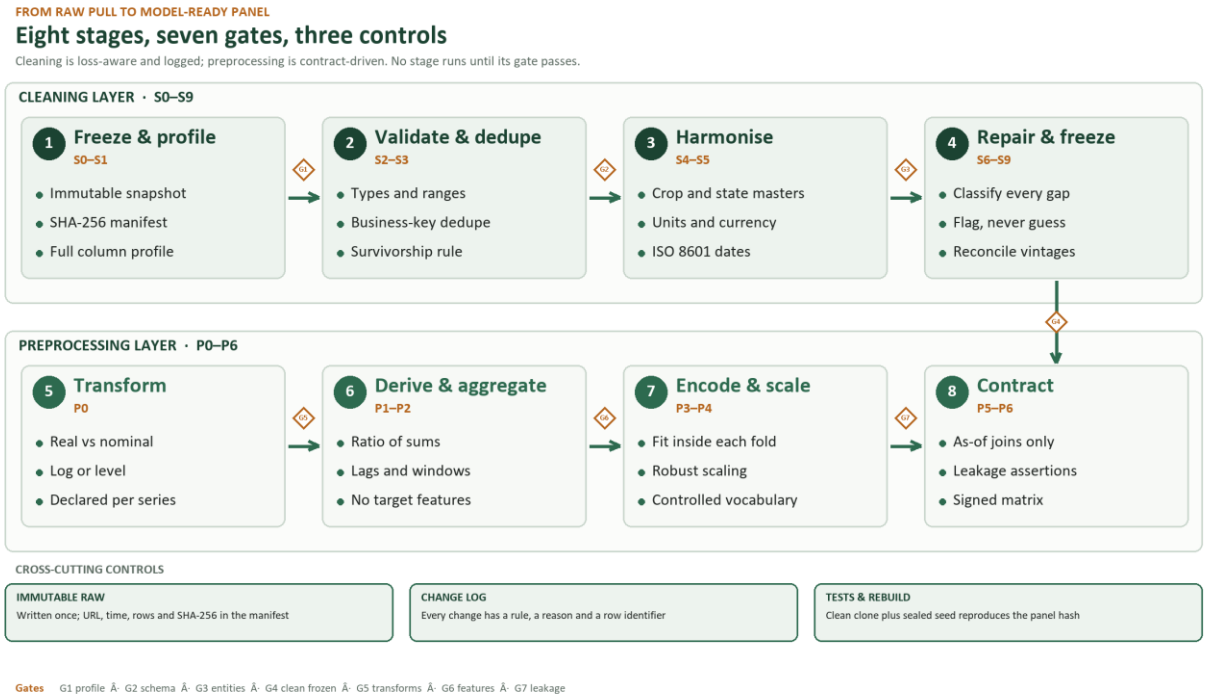


Figure 1. *The cleaning and preprocessing workflow. Gates G1–G4 govern the clean layer; G5–G7 govern the model-ready matrix.*

A diagram is useful for orientation and useless for execution. The numbered sequence below is the executable form: seventeen steps in the order they run, each with its output.

8.1 The seventeen-step sequence

Table 30. *The executable sequence. Steps 1–10 are cleaning; 11–17 are preprocessing.*

#	Step	Action	Output
1	S0	Snapshot the source; record URL, time, rows, licence and SHA-256	Immutable raw file + manifest row
2	S1	Profile columns, check row absence against the expected grid, open the defect register	Profile report + defect register
3	S2	Validate types, domains, enums, cross-field logic and referential integrity	Validated table + nulled invalid values
4	S3	Resolve exact, key and near duplicates with the survivorship rule	Unique business keys
5	S4	Map crops, states, districts and mandis; convert units and currency	Harmonised table
6	S5	Parse dates to ISO; derive crop, fiscal and marketing years; attach the trading calendar	Calendar-complete table
7	S6	Classify every gap by cause; interpolate short internal gaps; leave the rest	Clean price table with flags
8	S7	Screen outliers on log price; corroborate or flag; correct only mechanical errors	Outlier flags + corrections
9	S8	Apply source precedence; keep every vintage; build as-of views	Reconciled table + point-in-time view
10	S9	Run the acceptance suite, publish the scorecard, hash and freeze	Frozen clean layer
11	P0	Declare transformations per variable, with formula and availability rule	Transform registry

#	Step	Action	Output
12	P1	Derive indicators and lags; enforce the availability rule at creation	Indicator panel
13	P2	Aggregate with a ratio of sums; never a mean of ratios	Analysis panel
14	P3	Encode categorical variables inside the training fold	Encoded features
15	P4	Scale numeric features inside each fold, robustly where outliers remain	Scaled features
16	P5	Assemble the model matrix; attach the contract and the source hash	Model-ready matrix
17	P6	Run leakage, schema and reproducibility tests; fail the build on any breach	Signed matrix + test report

Two features of the sequence are easy to miss. Step 11 comes before any derivation, so the availability rule is declared rather than discovered. Step 17 can fail the build, which means the tests have authority; a test suite that only warns is a suggestion.

8.2 Time and ownership

Table 31. *Indicative effort for the first pass. The estimates assume the source register and masters already exist.*

Stage	Days	Owner	Reviewer
S0–S1 intake and profiling	1.0	Data engineer	Analyst
S2–S3 validation and deduplication	0.5	Data engineer	Analyst
S4–S5 harmonisation	1.0	Analyst	Domain reviewer
S6–S8 missing, outliers, reconciliation	1.5	Analyst	Statistician
S9 freeze and sign-off	0.5	Analyst	Project lead
P0–P6 preprocessing	2.0	Analyst	Statistician

The effort concentrates in S6–S8, which is where judgement is required. That is also the part that is easiest to rush and hardest to audit later. The review pairing is deliberate: the person who writes the imputation rule should not be the only person who signs it off.

9. Reproducibility

A cleaning decision that cannot be reproduced is an anecdote. Reproducibility is treated here as a set of mechanical controls, not as a virtue. Each control below prevents a specific failure, and each has a test that can be run without trusting the author.

9.1 Ten controls

Table 32. *Ten reproducibility controls. Each is cheap; the absence of any one is expensive.*

#	Control	Failure it prevents	Test
1	Immutable raw snapshots	A source is edited or silently refreshed	SHA-256 matches the manifest
2	Manifest with provenance	No one knows which pull produced a number	Every raw file has a row
3	Pinned environment	A library upgrade changes a result	pip install -r requirements.lock in CI
4	Sealed seeds	Stochastic steps differ between runs	Two runs produce identical hashes
5	One function per rule	A change in one place alters another	Unit tests per rule identifier
6	Cleaning log	The effect of cleaning is invisible	Every changed cell has a log row
7	Rule versioning	A rule changes without anyone noticing	Semver + changelog + scorecard diff
8	Run identifiers	Outputs from different runs are mixed	Every artefact carries run_id

#	Control	Failure it prevents	Test
9	Clean-clone rebuild	The pipeline only works on one machine	A fresh clone reproduces the hash
10	Retention policy	Old artefacts disappear or linger forever	Dated snapshots; documented retention

9.2 The manifest

The manifest is the spine of the audit trail. It is append-only, one row per retrieval, and it is the only place where a raw file's identity is recorded.

```
{
  "run_id": "2026-11-04T09:15:00Z-7f3a",
  "source_id": "AGMARKNET",
  "endpoint": "https://agmarknet.gov.in/...",
  "retrieved_at": "2026-11-04T09:15:00Z",
  "period_covered": "2020-01-01/2025-12-31",
  "rows": 3535,
  "sha256": "c0a070b878b301f5ca3ad37c402e7fb2...",
  "licence": "Government Open Data Licence - India",
  "status": "complete",
  "notes": "monthly modal price, state level; units as published"
}
```

Two fields do the most work. sha256 makes the snapshot verifiable; retrieved_at makes it possible to reconstruct which version of a revised series was in hand at a given time. Without the second, a point-in-time backtest is impossible even with perfect code.

9.3 Rule versioning

Rules change, and pretending otherwise produces a pipeline that is reproducible only by never improving it. The strategy versions rules with semantic versioning: a patch fixes an implementation without changing behaviour; a minor version adds a rule or a parameter; a major version changes an output. Every version carries a changelog entry and a scorecard diff against the previous run.

Table 33. *Rule versioning. A major change is allowed; an unrecorded one is not.*

Change	Version bump	Required evidence
Refactor, same output	Patch	Identical panel hash
New threshold or new rule	Minor	Scorecard diff + review
Output changes for existing cells	Major	Re-run, diff, re-sign-off, note in the release

9.4 The determinism test

Determinism is tested, not asserted. The procedure is deliberately blunt: delete the generated layers, re-run from the frozen raw snapshot with the sealed seed, and compare hashes. Any difference is a failure, including a difference that looks harmless.

- **Three hashes —** the clean price panel, the clean yield panel and the cleaning log are hashed after each run and compared with the previous run's values.
- **Fresh environment —** the run is repeated in a fresh virtual environment built from the lock file, on a different working directory, to catch hidden absolute paths and local state.
- **Negative control —** a deliberately perturbed input is fed through the suite. The tests must fail. A test suite that passes on corrupted input is not testing anything.

For the reference run, the clean price panel hashes to c0a070b878b301f5…, the yield panel to 49cbf0d0e6fa38c5…, and the cleaning log to be278d2ced2b5bc1…. These are the values a second run must reproduce. They are quoted in the report so that a reviewer can check the claim rather than take it on trust.

9.5 The clean-clone gate

The final gate is the one that matters to a handover: a person who has never seen the project clones the repository, installs from the lock file, runs one command, and gets the same panel. If that takes more than ten minutes, or requires a conversation, the pipeline is not finished. The gate is stated as a test rather than a hope because it is the only reproducibility claim that a reviewer can verify without reading the code.

> **REPRODUCIBILITY IS A PROPERTY OF THE DESIGN**
>
> If the raw data is immutable, the rules are functions, the log is complete and the environment is pinned, reproducibility follows. If any one of those is missing, no amount of documentation will recover it after the fact.

10. Worked Examples

The examples below are not illustrations written to fit the method. They are extracted from the reference run: the identifiers, old values, new values and flags are the pipeline's own log entries. Each case shows the raw state, the rule applied, the cleaned state and the downstream consequence of getting it wrong.

10.1 A mandi price in the wrong unit

The raw row is a cotton quote for Gujarat in 2024-02. It was filed as 91.579 under a per-quintal label. Cotton trades near ₹9,000 per quintal in this series, so the value is wrong by exactly two decimal places: it is a per-kilogram price in a per-quintal column.

Table 34. *E1: unit drift corrected by a factor of 100.*

Field	As received	After R06	Evidence
Crop / state	COTTON / Gujarat	COTTON / Gujarat	Unchanged
Period	2024-02-01	2024-02-01	ISO month
Modal price	91.579	9157.9	Ratio to series median ≈ 0.01
Flag	—	ok	Corrected, not merely flagged

```
R06  P02786  modal_price_inr_quintal  91.579 -> 9157.9
    action: corrected   reason: unit drift: INR/kg -> INR/quintal (x100)
```

The downstream consequence is easy to underestimate. Left uncorrected, the cell would lower the Gujarat cotton mean for that month by roughly 99% and the twelve-month mean by about 1.4%. A model would see a one-month crash and a partial recovery, and would learn a pattern that never happened. The correction is deterministic: the ratio is diagnostic, and no judgement is required. This is the only class of outlier the strategy overwrites.

10.2 Two names for one state, three formats for one month

A Haryana cotton row arrived with the state as 'Haryana (IND)' and a period that did not parse as ISO. The state string is a source convention, not a different state; the LGD code resolves it. The date is one of three formats present in the same column.

Table 35. *E2: entity and date harmonisation. The code is the key; the name is a label.*

Raw value	Canonical	Rule	Why it matters
Haryana (IND)	Haryana	R03	A join on the raw string silently drops Haryana
DD/MM/YYYY, e.g. 01/02/2021	2021-02-01	R04	ISO sort and join; no month/day swap
Mon-YYYY, e.g. Feb-2021	2021-02-01	R04	Same period, different representation
2021-02-01T00:00:00	2021-02-01	R04	ISO normalisation

The harmonisation is logged as a change of field 'state' from 'Haryana (IND)' to 'Haryana'. It looks trivial and it is not: in the raw panel there are several spellings for the same eight states, and a group-by on the raw column produces more than eight groups. Every aggregate after that point inherits the split, and no null count reveals it.

10.3 A missing month in a price series

COTTON in Gujarat has no quote for 2025-03. The months either side are observed at ₹9,321.04 and ₹8,136.40. The gap is internal and one month long, which places it in the only class eligible for imputation.

Table 36. *E3: a bounded imputation, with the checks that authorise it.*

Check	Value	Decision
Gap position	Internal (both anchors observed)	Eligible
Gap length	1 month	Within the ≤3-month bound
Cause	Not recorded; not structural	Treat as not reported
Anchor bound	9,321.04 – 8,136.40	Imputed value lies inside
Result	₹8,758.84	Time-weighted interpolation
Flag	imputed_short_gap	Value usable; provenance preserved

The imputed value is ₹8,758.84, not the arithmetic midpoint of the anchors. The interpolation is weighted by elapsed time, and February to April spans different month lengths. The difference is small here, but it is the kind of detail that makes a result reproducible rather than approximately right.

The flag is as important as the value. Every indicator that consumes this cell can be recomputed without it, and the sensitivity analysis in §6.1.4 does exactly that. Imputation is not hidden; it is labelled and reversible.

10.4 A festival spike and a digit error

Two values in the panel are extreme for opposite reasons. The first is a wheat quote of ₹4,066.53 in Rajasthan in 2022-02, during a documented supply disruption. The second is an onion quote of ₹71.02 in Madhya Pradesh, roughly two orders of magnitude below any plausible onion price.

Table 37. *E4: the same statistic, two different decisions. Corroboration is what separates an event from an error.*

Case	Value	Corroboration	Treatment	Flag
Wheat event	₹4,066.53	Neighbouring months and other markets elevated	Retain	spike_genuine
Onion error	₹71.02	No neighbour or market support; outside plausible band	Exclude from training; retain visibly	implausible

A z-score filter with a single threshold would delete both. Deleting the wheat event removes the observation the model most needs; deleting the onion error leaves a value that is obviously wrong but is at least visible. The strategy retains both and distinguishes them by flag, because the decision about what to exclude belongs to the modelling stage, not to the cleaning stage.

10.5 A unit mix-up in the yield table

The yield panel does not contain a yield error; it contains a production error and an area error that produce one. Both are detected by checking whether the implied yield is agronomically possible for the crop.

Table 38. *E5: two unit errors in the yield table, both recovered from the ratio of production to area.*

Case	Raw	Implied yield before	Correction	Yield after
WHEAT Punjab 2022	production 1,497,930 kt	4,042.2 t/ha	÷ 1,000 (tonnes → '000 t)	4.042 t/ha
MAIZE Haryana 2023	area 785.21 kha	1.33 t/ha	÷ 2.471 (acres → ha)	3.276 t/ha

Without the repair, the wheat record would show a yield near 4,042 t/ha and the maize record near 1.33 t/ha. Either would dominate any state or national average and would look like a spectacular harvest rather than a

filing error. The correction is justified because dividing by the factor restores the value to the crop's plausible range; a value that merely looks high, without a diagnostic factor, would be flagged instead.

The aggregation rule matters here too. State yield is computed as total production divided by total area, not as the mean of district yields. The ratio of sums gives large districts the weight their area deserves; a mean of ratios gives a 2,000-hectare district the same weight as a 200,000-hectare one.

10.6 A revised estimate and the forecast that could not have seen it

Wheat in Maharashtra, 2024, exists in three releases. Each is a correct record of what was published on its date; none is a correction of the others.

Table 39. *E6: three vintages of one fact. A point-in-time join selects the first row only for a July 2024 origin.*

Release	Published	Production (kt)	Yield (t/ha)	Knowable at 31 July 2024?
advance	2024-07-14	1,415.63	4.107	Yes
second	2024-10-01	1,387.32	4.025	No
revised	2025-02-15	1,436.86	4.169	No

Using the revised figure for a forecast made in July 2024 would improve the backtest by about 1.5% in yield terms, and the improvement would be entirely fictitious: the information did not exist at the origin. The as-of join in §6.5 prevents it with a single predicate. The cost of getting this wrong is not a wrong number in a table; it is a model that appears to work and fails in production.

10.7 What the six cases have in common

Table 40. *The examples are all reversible. That is the practical meaning of §4's provenance principle.*

Case	Defect class	Rule	Intervention	Reversible?
E1	Unit and scale	R06	Correct by factor	Yes — log holds the raw value
E2	Entity and format	R03/R04	Map and parse	Yes — mapping table
E3	Missing value	R08	Impute with a flag	Yes — flag and anchors retained
E4	Outlier	R07/R14	Flag; retain or exclude	Yes — value unchanged
E5	Unit and scale	R10	Correct by factor	Yes — log holds the raw value
E6	Vintage	R11	Select as of origin	Yes — every release retained

None of the six required a human to type a corrected value into a table. Each required a decision about a rule, made once, recorded, and applied consistently. That is the difference between cleaning a dataset and curating a result.

11. Quality Assurance and Acceptance

Assurance is the evidence that the strategy was followed, not a statement that it was. The section defines the scorecard, the acceptance criteria, the error budget and the sign-off record.

11.1 The quality scorecard

Table 41. *The scorecard. Each metric has a denominator, a target and a measured value.*

Metric	Definition	Target	Reference run
Completeness	Usable price cells ÷ expected grid	≥ 98%	98.5%
Duplicate rate	Duplicate business keys ÷ rows	0	0.0%
Entity conformance	Rows mapped to a canonical master ÷ rows	100%	100%
Date conformance	ISO dates ÷ rows	100%	100%
Imputation share	Imputed cells ÷ price cells	< 5%	4.5%
Unresolved flags	Flagged cells without a rule-level resolution	< 0.5%	0.1%
Log completeness	Changed cells with a log row ÷ changed cells	100%	100%
Rebuild time	Clean clone to identical hash	≤ 10 minutes	verified

The unresolved-flag rate is the one that most often fails in practice. In the reference run 3 cells remain flagged as unexplained, which is 0.1% of the panel — comfortably inside the 0.5% budget. A rate above the budget does not mean the data is unusable; it means the defect register has an open investigation that must be closed before sign-off.

11.2 The error budget

Cleaning is allowed to change the data, but not without limit. The error budget states how much change each step may cause before it requires escalation.

Table 42. *The error budget. A budget turns a judgement call into a threshold that triggers a review.*

Step	Budget	Escalation if exceeded
Deduplication	≤ 5% of rows	Investigate the source before dropping; a high duplicate rate usually means a re-pull loop
Harmonisation	≤ 10% of rows mapped	Review the master; a large mapping suggests a new source convention
Imputation	≤ 5% of cells	Drop the series from modelling; report the coverage gap
Outlier correction	≤ 1% of cells	Stop correcting; switch to flagging and review the unit table
Outlier exclusion	≤ 2% of cells	Review the threshold and the seasonal screen
Row loss overall	≤ 2% of rows	Written justification and project-lead sign-off

Budgets are not moral limits; they are attention limits. A step that changes 15% of rows may be entirely correct, but it is unusual enough to deserve a second pair of eyes before the result is used. The budget is the mechanism that buys those eyes.

11.3 Sign-off record

Table 43. *Sign-off. The reviewer who verifies the rebuild should not be the author of the pipeline.*

Check	Owner	Evidence required
Source register complete	Data engineer	Every series has a source, licence and refresh rate

Check	Owner	Evidence required
Masters current	Analyst	Crop, geography and unit masters dated within the quarter
Cleaning log complete	Analyst	Every changed cell has a rule and a reason
Scorecard passes	Analyst	All eight metrics at or inside target
Sensitivity run	Statistician	Three-way comparison for every indicator using imputed values
Point-in-time test	Statistician	No feature after origin; vintages selected as of date
Rebuild verified	Reviewer	Clean clone reproduces the panel hash

The sign-off is a record, not a ceremony. Each line names the evidence that exists, so a later reviewer can check the claim without reconstructing the project's history. If a line cannot be evidenced, it is not signed.

12. Risks, Delivery and Conclusion

12.1 Risks and controls

Table 44. *Eight risks, their controls and the mechanism that makes the control real.*

Risk	Likelihood	Impact	Control
Source format changes mid-project	High	Medium	Schema validation fails loudly; parser is versioned; re-pull is logged
Official series revised retrospectively	High	High	Vintage ids and as-of joins; every release retained
Geography codes change	Medium	High	Effective-dated mapping; LGD as key; name is a label
Over-cleaning removes genuine events	Medium	High	Flag before correcting; corroboration test; error budget
Imputation dominates a series	Medium	Medium	5% cap; sensitivity analysis; drop rather than invent
Rule drift between runs	Medium	Medium	Versioned rules; scorecard diff; CI
Key-person dependency	Medium	High	Clean-clone gate; documented runbook; paired review
Suppressed cells inferred	Low	High	Prohibition in the rule register; suppression flag; audit

12.2 Five-day delivery plan

Table 45. *Five working days from raw snapshot to signed matrix, assuming the source register exists.*

Day	Focus	Output	Gate
1	Masters, config, raw snapshot and profile	Source register, masters, profile report	G1
2	Validation, deduplication and harmonisation	Clean keys, canonical entities and units	G2–G3
3	Missing values, outliers and reconciliation	Clean tables with flags and log	G3–G4
4	Preprocessing framework and matrix	Indicator registry, model matrix, contract	G5–G6
5	Tests, scorecard, documentation and handover	Signed clean layer, README, runbook	G7

The plan is deliberately front-loaded. Days 1–2 establish the masters and the canonical shape; if they slip, every later step slips with them. Days 3–4 are the judgement-heavy part and the part most likely to surface a source problem that changes the scope. Day 5 is reserved for evidence, not for finishing the cleaning.

12.3 What would change on real data

The reference panel is regular and well behaved in ways that real sources are not. Four differences would matter, and the strategy anticipates them:

- **Irregular frequency —** daily mandi quotes arrive with gaps that are not rectangular. The grain and the gap rule change: the three-month bound becomes seven days, and the trading calendar becomes the primary explanation for zeros.
- **PDF-only tables —** some sources publish only PDFs or image maps. Extraction becomes a cleaning step in its own right, with the extracted value checked against a published total before it is trusted.
- **Suppression —** district yields are often withheld or reported with wide revisions. Suppression and vintage discipline stop being theoretical and become the dominant risk.
- **Unstable portals —** state portals change their column names and date conventions without notice. Schema validation is the control; a silent rename is caught at load, not in the model.

None of these requires a different strategy. They require the parameters in Appendix C to be reset for the frequency and the source, which is why the parameters are separated from the rules.

12.4 Conclusion

The strategy can be summarised as a sequence of refusals. It refuses to edit raw data, to impute without a cause, to delete a value because it is inconvenient, to average two disagreeing sources into a third number, and to call a pipeline reproducible because it ran once on one machine. Each refusal is implemented as a rule, a flag or a test, and each is visible in the artefacts the project hands over.

The reference run shows that the approach is practical at the scale of the project. It converts a panel with 92.3% usable price cells into one with 98.5%, resolves every duplicate key, harmonises every entity, and leaves 550 logged interventions that a reviewer can audit line by line. It does so without deleting a single genuine price event and without inventing a value it cannot bound.

That is the standard the later weeks inherit. The exploratory analysis in Week 3 can trust that its coverage figures mean what they say; the models in Week 4 can trust that their features were available at the origin; and the evaluation in Week 5 can trust that the backtest was not flattered by a revision the model could not have seen.

Appendix A. Cleaning-Log Schema

The cleaning log is the audit trail and a first-class deliverable. It is append-only, one row per changed, imputed, flagged or dropped value, and it joins back to the clean table on row_id and to the rule register on rule_id.

Table 46. *Cleaning-log schema. The reason field is prose so that a reviewer does not have to decode an enum.*

Field	Type	Example	Notes
step	enum	S6	Pipeline step that made the change
rule_id	text	R08	Foreign key to the rule register (Appendix C)
table	enum	prices	prices, yields or a derived table
row_id	text	P02799	Stable identifier from the raw load
field	text	modal_price_inr_quintal	The column changed
old_value	text	91.579	Raw value; empty when the cell was null
new_value	text	9157.9	Clean value; empty for drops and flags
action	enum	corrected	corrected, imputed, harmonised, parsed, flagged, dropped, nulled, derived, left_missing
reason	text	unit drift: INR/kg -> INR/quintal (x100)	Human-readable cause, not a code
run_id	text	2026-11-04T09:15Z-7f3a	Ties the row to a pipeline run

An extract from the reference run, with three different actions:

```
step rule  row_id   field              old_value    new_value   action
S4   R06   P02786   modal_price        91.579       9157.9      corrected
S6   R08   P02799   modal_price                     8758.84     imputed
S7   R07   P00530   modal_price        4066.53      4066.53     flagged

reason (third row): genuine spike corroborated by neighbours; retained
```

The full log for MYP-26 has 550 rows. It is small enough to review and large enough to answer the question every reviewer eventually asks: what did you change, and why?

Appendix B. Quality Flags and Dictionary Extension

Every fact row carries a quality_flag. The flag is never null, and it is the single field that determines whether a value enters a model. The vocabulary is closed; a new flag is a schema change, not a free-text annotation.

Table 47. *The closed flag vocabulary. The model-inclusion column is the operational meaning of each flag.*

Flag	Meaning	In the clean table	Used in models?
ok	Observed and valid	Value present	Yes
imputed_short_gap	Interpolated across ≤3 months	Value present	Yes, with sensitivity run
imputed_bounded	Imputed with an explicit domain bound	Value present	Yes, with sensitivity run
market_closed	Zero price reclassified as no trade	Null	No; calendar feature instead
invalid	Negative or impossible value	Null	No
missing_long_gap	Gap > 3 months	Null	No
suppressed	Withheld by the publisher	Null	No; never inferred
outlier_unresolved	Robust outlier without corroboration	Value present	No

Flag	Meaning	In the clean table	Used in models?
spike_genuine	Corroborated genuine event	Value present	Yes
implausible	Outside the physical domain after repair	Value present	No
provisional	Advance estimate; later releases expected	Value present	Only in point-in-time use
discontinued	Series ended; no current observation	Historical values only	No

The data dictionary is extended with five fields beyond the Week 1 specification: quality_flag, outlier_z, vintage_id, available_at and rule_version. The first two describe the value; the next two describe when it was knowable; the last describes the rule that produced it. Together they make every row self-describing. The extension is reproduced below so that the Week 1 dictionary and the implementation cannot drift apart.

Table 48. *Dictionary extension. The availability column is the field that prevents leakage when the indicator is used as a feature.*

Field	Type	Unit	Definition	Available when
modal_price_inr_quintal	float	INR / quintal	State modal wholesale price, canonical unit	Source publication
price_real_inr_quintal	float	INR 2011-12	Modal price deflated by CPI-IW	Deflator release
log_price	float	log INR	Natural log of the real price	Derived
arrival_tonnes	float	tonnes	Market arrivals, aggregated to state-month	Source publication
production_kt	float	'000 tonnes	Production estimate for the crop year	Vintage release
area_kha	float	'000 hectares	Area sown or harvested	Vintage release
yield_t_ha	float	t / ha	Production ÷ area, ratio of sums	Derived at vintage
yield_trend_dev	float	% of trend	Deviation from a five-year local trend	Needs five prior years
rainfall_anomaly_mm	float	mm	Departure from the 30-year local normal	Meteorological release
gdd_c	float	°C days	Growing-degree days above the crop base	Daily weather, lagged
crop_year	text	label	Growing and marketing year (e.g. 2024-25)	Derived from period
vintage_id	date	ISO date	Publication date of the release used	At load
available_at	timestamp	UTC	Earliest moment the value was knowable	At load
quality_flag	enum	—	Closed-vocabulary status controlling model inclusion	At clean
outlier_z	float	score	Modified z on log price within crop-state	At clean
rule_version	text	semver	Version of the rule that produced the row	At clean

Appendix C. Rule Register

The register is the contract between the strategy and the code. Each rule has an identifier, a parameter, a test and a record of whether it fired on the reference panel. A rule that does not fire is still tested; absence of the defect is not absence of the rule.

Table 49. *The rule register. 'Fired' is the number of log rows on MYP-26; 'by design' rules are structural and are tested rather than counted.*

ID	Rule	Key parameters	Test	Fired
R01	Schema and domain validation	Price > 0; production ≥ 0; area > 0	Invalid values are nulled, never clipped	47
R02	Duplicate resolution	Key = crop + state + period; survivorship order	Zero duplicate keys after the step	79
R03	Entity canonicalisation	Crop, state, district and mandi maps	Zero unmapped entities	67
R04	Date parsing	ISO, DD/MM/YYYY, Mon-YYYY; day-first convention	Zero unparsed dates; no ambiguous date guessed	117
R05	Calendar derivation	Crop, fiscal and marketing year definitions	Every row has all three years	by design
R06	Unit and scale repair	Ratio bands 0.004–0.025, 0.05–0.2, 5–20	Corrected values inside the plausible band	62
R07	Robust outlier flagging	Log price; MAD; |z| > 3.0; neighbour corroboration	Every flag has a reason and a review status	6
R08	Missing classification and bounded imputation	≤3 months; internal only; three bounds	No imputed target, suppression or long gap	160
R09	Ratio-of-sums aggregation	Σproduction ÷ Σarea	State yield equals total ÷ total, not mean of ratios	by design
R10	Derived-quantity unit repair	Yield > 8× or < 0.55× crop median	Repaired yield inside the crop range	10
R11	Vintage separation	One row per release; available_at	No release overwritten	by design
R12	Source reconciliation	Precedence per indicator; 5% tolerance	Disagreements recorded, never averaged	by design
R13	Suppression handling	Flag only; no inference	No suppressed cell has a value	not triggered
R14	Plausibility bounds	400–20,000 INR/quintal; crop yield ranges	No unflagged impossible value	2
R15	As-of join	available_at <= origin_date	No feature timestamp after its origin	by design
R16	Fold-local scaling	Fit on train only; robust where outliers retained	Scaler statistics differ by fold	by design
R17	Fold-local encoding	One-hot ≤10 levels; target encoding above	No global target encoding	by design
R18	Output contract	Schema, units, hash, dictionary version	Contract test passes before release	by design

The parameters are deliberately separated from the rules. A weekly panel changes the three-month gap bound to seven days without changing the logic of R08; a new crop changes the plausible band without changing R14. That separation is what makes the register reusable across Weeks 3–5 and across datasets.

Appendix D. Reproducibility Checklist

Run this list before any handover. Each item is binary; there is no partial credit.

Table 50. *Twelve checks. Items 9 and 10 are the ones most often skipped and most informative.*

#	Check	Evidence
1	Raw snapshots are immutable and hash-recorded	Manifest with SHA-256 per file
2	Every raw file has a source, licence and retrieval time	Manifest row completeness = 100%

#	Check	Evidence
3	Environment is pinned	requirements.lock; Python version recorded
4	All stochastic steps have a sealed seed	Seed in config and manifest
5	Every rule is a function with a unit test	Test suite passes
6	Every changed value is in the cleaning log	Log count reconciliation
7	Rules are versioned with a changelog	Semver and scorecard diff
8	Every artefact carries a run_id	Artefact headers and filenames
9	A clean clone reproduces the panel hash	Hash comparison, not a visual check
10	A negative control fails as expected	Perturbed input trips the suite
11	The data dictionary matches the schema	Dictionary validation test
12	The README runs end to end in ≤10 minutes	Timed on a clean machine

THE TEN-MINUTE TEST

Hand the repository to a colleague who has not seen it, with no verbal instructions. If they cannot reproduce the panel in ten minutes, the documentation is part of the bug.

Appendix E. Glossary

Table 51. *Terms as they are used in this document, not as they might be used elsewhere.*

Term	Definition
As-of join	A join that selects the version of a fact knowable at a given date, not the latest version.
Business key	The columns that uniquely identify a fact: crop, state and period for a monthly price.
Cleaning log	The append-only record of every changed, imputed, flagged or dropped value.
Crop year	The year of the growing and marketing cycle, which may not align with the calendar year.
Defect register	The prioritised list of data problems found in profiling, with a rule and an owner.
Imputation	Replacing a missing value with an estimate, with a flag and a stated bound.
LGD code	Local Government Directory code; the stable identifier for Indian states and districts.
MAD	Median absolute deviation; a robust measure of spread used in the modified z-score.
Mandi	A regulated wholesale market for agricultural produce.
MAR / MCAR / MNAR	Missingness mechanisms: at random, completely at random, and not at random.
Point-in-time	A view that uses only information available at the forecast origin.
Quality flag	The closed-vocabulary status attached to every fact row, controlling model inclusion.
Ratio of sums	An aggregate computed as Σnumerator ÷ Σdenominator, not as a mean of ratios.
Survivorship rule	The deterministic rule that decides which duplicate row is kept.
Vintage	A specific release of a series, identified by its publication date.
Winsorising	Replacing extreme values with a percentile; deliberately not used for prices here.
Quarterly / monthly modal price	The most frequently occurring price in a period, as reported by a market.
Leakage	Use of information in training that would not have been available at prediction time.

Appendix F. Assumptions, Limitations and Brief Compliance

F.1 Assumptions

- **A1 —** the source register and the four masters exist before cleaning begins. This document specifies them but does not build them.
- **A2 —** a monthly grain is sufficient for the price analysis; daily quotes are aggregated upstream with the same rules.
- **A3 —** the CPI deflator and the RBI reference rate are available for every period in scope.
- **A4 —** the plausible price and yield bands in Appendix C are wide enough to absorb normal seasonal and regional variation. They are parameters, not constants.

F.2 Limitations

Three limitations should be stated before the strategy is applied to real data. First, the three-month imputation bound is calibrated for monthly data and must be reset for weekly or daily series; applied unchanged to daily quotes it would interpolate across a quarter and smooth away exactly the variation of interest. Second, the univariate outlier screen has limited power on strongly seasonal series: two of seven planted digit errors survived it in the reference run, which is why the contextual screen is specified rather than optional. Third, the strategy cannot recover a value that is missing because the publisher suppressed it, and it should not try; the correct output is a flag and a coverage gap, not an estimate.

A fourth limitation is organisational rather than statistical. The controls only work if the project treats the rule register as a contract. A team that edits the clean table directly can satisfy every test in this document and still produce an unauditable result.

F.3 Compliance with the brief

Table 52. *How each element of the brief maps to the document.*

Requirement	Where it is addressed	Evidence
Introduction to common agribusiness data-quality issues	Section 3	Eight defect classes; six structural causes; impact analysis
Missing values, inconsistent formats and outliers	Sections 3, 5, 6	Taxonomy, validation, robust screens, worked cases
Specific methods — imputation, normalisation, filtering	Section 6, 7	Seven-method comparison; transformation table; bounds
Step-by-step cleaning plan	Section 5	S0–S9 with decision rules, outputs and gates
Proposed preprocessing framework	Section 7	Five layers, contracts, leakage tests
Workflow diagram or list	Section 8	Diagram plus a seventeen-step executable sequence
Importance of reproducibility	Section 9	Ten controls, manifest, determinism test, clean-clone gate
Hypothetical examples (bonus)	Section 10	Six cases extracted from a seeded reference run

Appendix G. Statistical Notes

These notes give the derivations behind the methods in Section 6. They are written for a reader who wants to check the reasoning or adapt the parameters, not as a statistics primer.

G.1 Modified z-score and the MAD

For observations $x_1 \ldots x_n$, let m = median(x) and MAD = median($|x_i - m|$). The modified z-score is

$$z_i = 0.6745 \times (x_i - m) / \text{MAD}$$

The constant 0.6745 makes the scale comparable to a standard deviation under normality, because for a normal distribution MAD ≈ 0.6745σ. The influence function of the median and the MAD is bounded: a single value can move the estimate only slightly, which is why the screen does not mask the very outlier it is testing. With a mean and standard deviation, one large error inflates σ and can push its own z-score below the threshold.

The price screen applies the statistic to log price. If p is log-normal, log p is normal, so the threshold has a stable interpretation across crops whose price levels differ by an order of magnitude. On the raw scale, a ₹500 deviation is large for onion and trivial for cotton; on the log scale, the same proportional deviation has the same score.

G.2 IQR fences

The distributional cross-check uses Q1 and Q3 and the fences Q1 − 1.5·IQR and Q3 + 1.5·IQR, with 3·IQR as the extreme fence. The fences make no distributional assumption, which is their advantage and their weakness: on a seasonal series the interquartile range is wide, so the fences are wide, and a moderate error can sit comfortably inside them. The IQR is a cross-check, not the decision rule.

G.3 Interpolation and its error

Linear interpolation between anchors (t_0, y_0) and (t_1, y_1) gives, for t between them,

$$\hat{y}(t) = y_0 + (y_1 - y_0) \times (t - t_0) / (t_1 - t_0)$$

Time-weighted interpolation uses the actual elapsed time, not the row index; the reference run's imputed March value is not the arithmetic midpoint because February to April spans different month lengths. The interpolation error is bounded by the curvature of the series over the gap: if the second derivative is bounded by M, the error is at most $M \cdot h^2/8$ for a gap of length h. That bound is why the rule limits the gap to three months: the error grows with the square of the gap, so a six-month gap is four times as uncertain as a three-month one.

Interpolation assumes the series is smooth across the gap. If a policy announcement, a festival or a market closure falls inside it, the assumption fails and no amount of neighbour information can recover the shape. The flag makes the assumption visible; the sensitivity analysis measures its effect.

G.4 MICE, in one paragraph

Multiple Imputation by Chained Equations fits a conditional model for each variable with missing data, using the other variables as predictors, and iterates until the imputations stabilise. It preserves cross-variable structure that interpolation ignores, and it produces multiple completed datasets whose variation reflects imputation uncertainty. It is the right tool when many indicators are correlated and the missingness is MAR. It is the wrong tool for a short gap in a single series, where it adds model dependence and runtime without adding information, and it is not a substitute for classifying the cause of the gap.

G.5 Variance-stabilising transforms

The Box-Cox family maps a positive variable to y(λ) = (x^λ − 1)/λ for λ ≠ 0 and log x for λ = 0. The Yeo-Johnson extension handles zero and negative values. For prices the log (λ = 0) is the default because it has a direct interpretation as a proportional change and because the series is strictly positive after validation. The transformation is applied before outlier screening and before modelling, and it is inverted before any result is reported in rupees.

G.6 STL residuals

STL decomposes a series into trend, seasonal and remainder components by local regression. The remainder is the part not explained by level or season, and it is the natural input to a contextual outlier screen: a value is surprising if its remainder is extreme, not if its level is. STL requires at least two full seasonal cycles, which is why it is a second-pass screen on the longer series rather than a first-pass rule on every crop–state pair.

G.7 Mahalanobis distance and isolation forests

The Mahalanobis distance $d^2 = (x - \mu)^T \Sigma^{-1} (x - \mu)$ measures how far a multivariate observation is from the centre, accounting for covariance. It is powerful when indicators are correlated (price, arrivals, rainfall) and misleading when Σ is estimated from the same contaminated sample. An isolation forest isolates observations by random partitioning and scores them by how quickly they separate; it scales well and makes few assumptions, but its output is a score, not a cause. Neither method replaces the corroboration test: a multivariate flag still needs an explanation before a value is excluded.

G.8 Why a ratio of sums

Let districts i have production P_i and area A_i. The state yield is $\Sigma P_i / \Sigma A_i$. The mean of district yields is $(1/n)\cdot\Sigma(P_i/A_i)$, which weights a small district equally with a large one. The two coincide only when all areas are equal. Since area shares are highly unequal, the mean of ratios is biased toward the yields of small districts, and the bias does not vanish as n grows. The ratio of sums is the only aggregate that preserves the physical identity production = yield × area.

Appendix H. Test Catalogue

The tests below run in CI on every build. A failing test blocks the release; a warning is not a control. The catalogue is deliberately short enough to be read in one sitting.

Table 53. *Fifteen tests. T14 is the one that keeps the other fourteen honest.*

ID	Test	Assertion	Failure means
T01	Manifest integrity	Every raw file has a row and a matching SHA-256	The evidence base is incomplete
T02	Key uniqueness	Zero duplicate business keys in L2	Deduplication failed or the key is wrong
T03	Completeness	Usable price cells ÷ grid ≥ 98%	A series has a coverage gap
T04	Type and domain	No invalid value without a flag	Validation was bypassed
T05	Entity conformance	100% of rows map to a master	A new alias was introduced
T06	Date conformance	100% ISO month starts	A parser silently failed
T07	Imputation bounds	No imputed target, suppression or long gap	R08 was applied outside its scope
T08	Flag completeness	No null flag; no unknown flag	A new state was introduced without a schema change
T09	Log reconciliation	Changed cells = log rows for changes	A change was made outside the rules
T10	Point-in-time	No feature available_at after origin	Vintage leakage
T11	Fold-local transforms	Scaler and encoder statistics differ by fold	Global fitting leaked the test period
T12	Target isolation	No feature is a deterministic function of the target	Target leakage
T13	Determinism	Two runs produce identical hashes	A hidden source of randomness or state
T14	Negative control	Perturbed input fails the suite	The tests are not testing
T15	Contract	Schema, units and hash match the declaration	The matrix is not the one that was signed